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Boat Cover Device

Abstract

An air cover device is disclosed for easily inflating, setting up, and removing a boat cover. The inflatable boat air cover device comprises a body component that is configured in a variety of shapes and includes a frame designed from a continuous pressurized air tube system. The device also comprises a plurality of individual inflatable tubes inserted into sleeves of the device for enhanced support. The boat cover can be attached and removed by one individual. The cover is secured to a boat or under a dry dock vessel via plurality of fasteners, such as but not limited to straps and snaps. Further, the device can be inflated and deflated via a battery-operated portable air compressor for ease and convenience.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to, and the benefit of, U.S. Provisional Application No. 63/559,492, which was filed on Feb. 29, 2024, and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates generally to the field of boat cover devices. More specifically, the present invention relates to an improved air cover for pontoon boats, runabouts, deck boats, or suitable boat. Accordingly, the present disclosure makes specific reference thereto. Nonetheless, it is to be appreciated that aspects of the present invention are also equally applicable to other like applications, devices, and methods of manufacture.

BACKGROUND

[0003] By way of background, this invention relates to improvements in boat cover devices. Generally, individuals owning pontoon boats, runabouts, deck boats or similar boats lack the means to properly secure and cover the boat to prevent potential damage. Further, standard boat covers are typically designed with an independent pole system that can be cumbersome to set up each time a boat needs covered or secured. Moreover, standard boat covers may not be able to properly accommodate pooling of rainwater, creating sags that require users to drain the cover manually. Users need a boat cover device that is collapsible, easy to set up and store, while providing an easier way to drain rainwater.

[0004] Further, the improved boat cover device is portable, easily set up, and collapsible for easy transportation and storage. Often, boat covers are designed with a metal pole system to adequately support and protect a boat that requires more than one person to set up. Without an inflatable boat cover or similar device, the user is unable to independently install the boat cover. Specifically, the inflatable boat cover device enables a single person to install the device in preparation to protect and secure their boat against strong winds and potential damage. Finally, boat covers with a metal pole system often collapse when exposed to high winds, whereas the inflatable boat cover is more durable and can withstand stronger winds with an added strap or snap fastener system.

[0005] Accordingly, there is a demand for an improved boat air cover device that provides protection for pontoon boats, runabouts, deck boats or similar watercrafts from damage caused by water pooling, rainwater, or strong winds. More particularly, there is a demand for a boat cover device that can be installed by a single person, easily inflatable and collapsible for simple transportation and storage.

[0006] Therefore, there exists a long felt need in the art for a boat cover device that provides users with an inflatable boat cover. There is also a long felt need in the art for an inflatable boat cover that comprises a portable air compressor. Further, there is a long felt need in the art for an air cover device that has a rechargeable air compressor. Moreover, there is a long felt need in the art for a device that has inflatable air bladders for added support, rather than a metal pole system. Further, there is a long felt need in the art for a boat cover device that can be inflated and installed by one person. Moreover, there is a long felt need in the art for a boat cover device that is collapsible for easy transportation and storage. Finally, there is a long felt need in the art for an air cover device that has a plurality of fasteners to secure to a boat or dry dock vessel for optimal support.

[0007] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises an air cover device. The device is a multipurpose, inflatable boat cover device that provides easy installation and removal for one individual. The air cover device comprises a body component that is configured in a variety of designs and comprised of a frame manufactured from a durable fabric. Preferably, the frame is manufactured from a durable fabric, such as but not limited to a marine grade vinyl material that comprises a plurality of air tubes or bladders for support. Further, the air

cover is secured to the boat via straps or snaps during use. The air can be inflated, removed, and deflated to a collapsed state for storage and transportation. Further, the air cover comprises a battery-operated portable air compressor which is rechargeable.

[0008] In this manner, the air cover device of the present invention accomplishes all of the forgoing objectives and provides users with a device that provides an inflatable air cover for boats. The device has a rechargeable battery-operated portable air compressor. The device can be collapsed for storage/transport.

SUMMARY OF THE INVENTION

[0009] The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to present some general concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0010] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises an air cover device. The device is a multipurpose, inflatable air cover for boats that is easily blown up and transported. The air cover device comprises a body component with a frame constructed from a durable fabric that. The device frame includes a plurality of air tubes or bladders for enhanced support. The body component comprises a plurality of fasteners around the perimeter of the device for securing the cover to the boat or under a dry dock vessel when stored or in use during a rainstorm. Further, the air cover comprises a battery-operated portable air compressor which is rechargeable.

[0011] The present invention is directed to a boat air cover device that has particular utility in protecting watercrafts, such as pontoon boats, roundabouts, deck boats, and other similar boats from damage caused by rainstorms or high winds. In the present invention, the air cover device can be inflated and collapsed easily by one individual. Although the device can be manually inflated, the device can be inflated using a rechargeable battery-operated air compressor. An individual user can attach the device to a pontoon boat or similar boat using at least one fastener, inflate the device, deflate the device, and collapse the device with ease for transportation and storage.

[0012] In one embodiment, the air cover device is adapted for an individual to independently cover a pontoon boat, roundabout, or similar boat as is known in the art. Generally, the boat cover device comprises a body component, including a fabric frame with a plurality of air tubes or a continuous air tube system.

[0013] In one embodiment, the base component is typically manufactured with a heavy-duty material, such as marine grade vinyl, but may also be manufactured from any suitable durable fabric, as well. Preferably, the fabric is waterproof and mildew resistant. In one embodiment, the body component is comprised of a continuous connected pressurized air tube system with a single inflatable point. Generally, the continuous air tube system provides the frame and support for the inflatable air cover device to protect boats from rainwater, hailstorms, snow, freezing temperatures, or other damaging weather conditions. Further, the continuous air tube system utilizes a single point of inflation, enabling the device to be secured and inflated by a single person for ease and convenience.

[0014] In another embodiment, the base component comprises individual inflatable tubes, each with a threaded valve for inflation. Further, the inflatable tubes are inserted into sleeves on the interior side of the air cover device, creating a sturdy structured frame. Moreover, the inflatable tubes are interconnected via fasteners such as but not limited to snaps and straps for increased stability. Thus, the vinyl sleeves or air tubes can be inserted and removed easily for repair, replacement, or storage. Moreover, the frame can be expanded into an elongated configuration during use or collapsed for storage. In other words, in the collapsed mode, the size and/or the volume occupied by the device is significantly reduced, thereby facilitating efficient storage and ease of transport. Further, the frame can be manufactured in a variety of sizes and configurations to

accommodate a variety of boat sizes and types.

[0015] In one embodiment, the air cover device comprises an opening on one side. Typically, once the air cover is inflated and secured to a boat or dry vessel dock, users can enter and exit the device through the opening. Further, users can close the opening using a closure system such as but not limited to a zipper. Moreover, a zipper enables users to easily go into the boat or watercraft under the cover to ensure that the perimeter of the device is properly secured around the interior and exterior of the boat. Finally, a zipper system is easy and convenient for users of all ages.

[0016] In one embodiment, the air cover device comprises at least one valve to inflate or collapse the boat cover. Typically, the valve is threaded and manufactured from a durable plastic or durable rust proof metal material. Further, a threaded valve provides airtight security to ensure the inflatable cover does not deflate unintentionally, placing the boat or watercraft at risk for damage. Finally, the threaded valve may be accessed from the interior or exterior of the air cover device for convenience.

[0017] In one embodiment, the air cover device comprises a port for a handheld pump for inflating. Typically, the device will be inflated via the rechargeable air compressor, however, the device provides an alternative to accommodate a variety of situations.

[0018] In one embodiment, the air cover device comprises a portable air compressor for easy set up and removal. Further, the air compressor is battery powered. Moreover, the battery is a rechargeable battery and is typically a USB rechargeable battery to provide power to the air compressor. Moreover, the air compressor comprises a switch to inflate or collapse the device. Specifically, when deflating the device, attaching the air compressor to the device valve and turning the switch to deflate removes the air exponentially faster than enabling the air to leave naturally. Finally, the air compressor can inflate and collapse the device quickly for easy storage and transportation.

[0019] In one embodiment, the base component is secured to the boat rail or under a dry dock vessel via fasteners. Typically, the boat air cover is secured to a rail or dry dock vessel via straps, snaps, or other suitable securing means as is known in the art. In use, the air tube system or individual tubes are inflated to expand the frame and secure the device via fasteners to a pontoon boat, roundabout, or other watercraft. When a user wants to collapse the device, at least one valve is opened, air is released via the air compressor or naturally, and the frame is collapsed for storage.

[0020] In yet another embodiment, the air cover device comprises a plurality of indicia. Typically, the device is manufactured in a neutral color such as but not limited to beige. Further, the cover can be manufactured in a variety of colors to meet the users' desires.

[0021] In yet another embodiment, a method of using and storing the device. The method includes the steps of providing an inflatable air cover device comprising a body component with an air system frame and a portable air compressor. The method also comprises securing the air cover to boat rail or dock vessel via straps or snaps. Further, the method comprises inflating the device via the portable air compressor. The method also comprises deflating the device utilizing the battery-operated air compressor. Finally, the method comprises collapsing the device for storage and transportation.

[0022] Numerous benefits and advantages of this invention will become apparent to those skilled in the art to which it pertains, upon reading and understanding the following detailed specification.

[0023] To the accomplishment of the foregoing and related ends, certain illustrative aspects of the disclosed innovation are described herein in connection with the following description and the annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and are intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:

[0025] FIG. 1 illustrates a perspective view of one embodiment of the air cover device of the present invention in accordance with the disclosed architecture;

[0026] FIG. 2 illustrates a perspective view of one embodiment of the air cover device of the present invention showing the interior of the fabric insert in accordance with the disclosed architecture;

[0027] FIG. 3 illustrates a perspective view of one embodiment of the air cover device of the present invention showing the pockets of the fabric insert in accordance with the disclosed architecture;

[0028] FIG. 4 illustrates a flowchart showing the method of easily transporting supplies and gear on a beach in accordance with the disclosed architecture.

DETAILED DESCRIPTION OF THE PRESENT INVENTION

[0029] The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They are not intended as an exhaustive description of the invention and do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

[0030] As noted above, there is a long felt need in the art for an air cover device that provides users with an inflatable cover for boats. There is also a long felt need in the art for an air cover device that comprises a portable air compressor. Further, there is a long felt need in the art for an inflatable boat cover device that has a rechargeable battery. Moreover, there is a long felt need in the art for a device that has air tubes or bladders for support. Further, there is a long felt need in the art for an inflatable air cover device that is collapsible for easy transportation and storage. Finally, there is a long felt need in the art for an air cover device that has a plurality of fasteners for secure installation and optimal support.

[0031] The present invention, in one exemplary embodiment, is a novel boat cover device. The inflatable boat cover device comprises a base component that is configured in a variety of shapes and comprised of a frame designed from a continuous pressurized air tube system. The base component is preferably manufactured from a marine grade vinyl material with fasteners along the perimeter of the device for securing the cover to pontoon boats, roundabouts, or similar boats. Specifically, the air cover is secured to a boat or dock via straps or snaps during use. The device is portable, inflatable, and deflatable with ease, and can be secured to a boat or dock by a single person. Users can inflate or deflate the device using the portable rechargeable air compressor. The device provides enhanced support and protection to boats during inclement weather such as but not limited to, rain or hailstorms, freezing temperatures, etc. Moreover, the air cover protects boats from debris such as but not limited to bird droppings, leaves, twigs, and other airborne particles. The airtight threaded valve ensures the air cover device remains inflated and withstands a variety of weather conditions in addition to preventing unwanted pests, such as mice or other rodents from getting into the boat and causing damage. Further, the device comprises an opening for users that

closes via a zipper.

[0032] The present invention also includes a novel method of easily using, transporting, and storing the device. The method includes the steps of providing an inflatable air cover device comprising a body component with an air system frame and a portable air compressor. The method also comprises securing the air cover to a boat rail or under a dry dock vessel via straps or snaps. Further, the method comprises inflating the device using the portable air compressor via the threaded valve. The method also comprises deflating the device utilizing the battery-operated air compressor and collapsing the device. Finally, the method comprises detaching the device using the fasteners and storing the device.

[0033] Referring initially to the drawings, FIGS. 1-2 illustrate a perspective view of one embodiment of the inflatable air cover device **100** of the present invention. In the present embodiment, the inflatable boat air cover device **100** is an improved boat cover device **100** that provides a user with a multipurpose, inflatable boat cover device **100** that can be easily installed and removed by a single person. Specifically, air cover device **100** comprises a body component **102** that is configured in a variety of designs and manufactured from a durable fabric. Preferably, the body component **102** of the device **100** is manufactured from a durable fabric, such as but not limited to a marine grade vinyl material that is waterproof and mildew resistant. Further, the device **100** comprises a continuous air system **104** that provides the frame and enhanced support for the air cover device **100**. Specifically, the device **100** can be manufactured with a plurality of removeable interlocking air tubes **106** for increased support. Moreover, the continuous air system **104** and the inflatable tubes **106** provide the frame for the body component **102** and can be manufactured in a variety of configurations. Even further, the air system **104** and inflatable tubes **106** each have closeable threaded valves **114** to maintain an inflated state for the air cover device **100** while in use.

[0034] Generally, the air cover device **100** is attached to the rails of a pontoon boat, roundabout, suitable boat, or dry dock vessel using exterior fasteners **108** such as but not limited to straps and snaps. Further, the device **100** of the present invention can be secured to a boat **10** or dry dock vessel by a single person for ease and convenience. Moreover, the air cover device **100** comprises an opening **110** on one side surface of the device **100** to enable access to the boat **10** while the air cover **100** is in use. Specifically, the opening **110** can be fastened or unfastened using a zipper **112** for convenience. Typically, users of the device **100** can secure additional boating apparatus or equipment on the interior or under the air cover device **100** for increased protection. Specifically, the air cover device **100** protects the boat **10** from UV rays that can fade paint and tarnish fittings.

[0035] Further, the air cover device **100** is inflated using the portable air compressor **116**. Specifically, the portable air compressor **116** is power operated with a rechargeable battery **118** for ease of use and convenience for the user. Typically, users will inflate the device **100** using the portable air compressor **116**, however, the device **100** can be inflated using a handheld pump as well. Further, the air system **104** and inflatable tubes **106** each have at least one airtight closeable threaded valve **114** by which the air compressor **116** will fill the tubes **106** or air system **104** to a user's desired level. Thus, the air can be removed, and the air cover deflates to a collapsed state for storage and transport.

[0036] FIG. 3 illustrates a perspective view the air cover device **100** from the interior. In the present embodiment, the air cover device **100** can be manufactured from a durable vinyl material, preferably marine grade. The body component **102** of the device **100** includes a continuous air system **104** or inflatable tubes **106** to provide a frame and stability for the air cover device **100**. The air system **104** or inflatable tubes **106** span across the device **100**, from one side to the opposite side for increased support and protection for the boat **10**. Moreover, the opening **110** enables users access to the boat **10** when the device **100** is expanded and in use. Specifically, the zipper **112** enables users access for convenience. Moreover, users can utilize the opening **110** from the interior or exterior of the device **100** via the zipper **112**. Further, the air cover device **100** is comprised of interior fasteners **109** for enhanced protection during various types of damaging weather.

[0037] In one embodiment, the body component **102** of the device **100** comprises a plurality of indicia **120**. The base component **102** of the device **100** may include advertising, a trademark, or other letters, designs, or characters, printed, painted, stamped, or integrated into the body component **102**, or any other indicia **120** as is known in the art. Specifically, any suitable indicia **120** as is known in the art can be included, such as but not limited to, patterns, logos, emblems, images, symbols, designs, letters, words, characters, animals, advertisements, brands, etc., that may or may not be air cover, boat, or brand related.

[0038] FIG. **4** illustrates a flowchart of the method of easily securing and inflating the air cover to a boat or dry dock vessel. The method includes the steps of at **400**, providing an inflatable air cover device comprising a body component and a portable air compressor. The method also comprises at **402**, securing the air cover to a boat or under a dry dock vessel via fasteners. Further, the method comprises at **404**, inflating the device using the portable air compressor via at least one threaded valve. The method also comprises at **406**, deflating and collapsing the device using the battery-operated air compressor. Finally, the method comprises at **408**, removing the device by disengaging the fasteners from the boat or dock for storage.

[0039] Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different users may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not structure or function. As used herein “air cover device”, “inflatable air cover device”, “inflatable boat cover device”, and “device” are interchangeable and refer to the air cover device **100** of the present invention.

[0040] Notwithstanding the forgoing, the air cover device **100** of the present invention can be of any suitable size and configuration as is known in the art without affecting the overall concept of the invention, provided that it accomplishes the above stated objectives. One of ordinary skill in the art will appreciate that the air cover device **100** as shown in FIGS. **1-4** is for illustrative purposes only, and that many other sizes and shapes of the air cover device **100** are well within the scope of the present disclosure. Although the dimensions of the air over device **100** are important design parameters for user convenience, the inflatable air cover device **100** may be of any size that ensures optimal performance during use and/or that suits the user's needs and/or preferences.

[0041] Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. While the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features.

Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

[0042] What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are possible. Accordingly, the claimed subject matter is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims.

Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

Claims

1. An inflatable boat cover device comprising: a body component; a continuous air system; a plurality of inflatable air tubes; a frame; and a plurality of fasteners; wherein each of said plurality

of inflatable air tubes having a closeable threaded valve for inflation and deflation; wherein each of said plurality of inflatable air tubes are selectively removable; wherein said body component having a material selected from the group consisting of a marine grade vinyl material; wherein said frame comprising said plurality of inflatable air tubes and said continuous air system for supporting said body component; and further wherein said plurality of fasteners are a first series of exterior fasteners for securing said inflatable boat cover to a boat.

2. The inflatable boat cover device of claim 1, wherein said plurality of inflatable air tubes are interlocking.

3. The inflatable boat cover device of claim 2, wherein said material of said body component is waterproof.

4. The inflatable boat cover device of claim 3, wherein said material of said body component is mildew resistant.

5. The inflatable boat cover device of claim 2, wherein said plurality of fasteners are a second series of interior fasteners for securing said inflatable boat cover to the boat.

6. The inflatable boat cover device of claim 2, wherein said body component having a selectively openable door.

7. The inflatable boat cover device of claim 6, wherein said selectively openable door having a zipper for selectively fastening and unfastening said selectively openable door.

8. The inflatable boat cover device of claim 2, wherein said continuous air system is a portable air compressor.

9. The inflatable boat cover device of claim 2, wherein each of said plurality of inflatable air tubes span across an outer side of said body component from a first edge to an opposing second edge.

10. An inflatable boat cover device comprising: a body component; a continuous air system; a plurality of inflatable air tubes; a frame; and a plurality of fasteners; wherein each of said plurality of inflatable air tubes having a closeable threaded valve for inflation and deflation; wherein each of said plurality of inflatable air tubes are selectively removable; wherein said frame comprising said plurality of inflatable air tubes and said continuous air system for supporting said body component; wherein said plurality of fasteners are a first series of exterior fasteners for securing said inflatable boat cover to a boat; wherein said plurality of inflatable air tubes are interlocking; and further wherein each of said plurality of inflatable air tubes span across an outer side of said body component from a first edge to an opposing second edge.

11. The inflatable boat cover device of claim 10, wherein said body component having a material selected from the group consisting of a marine grade vinyl material, and further wherein said material of said body component is waterproof.

12. The inflatable boat cover device of claim 10, wherein said body component having a material selected from the group consisting of a marine grade vinyl material, and further wherein said material of said body component is mildew resistant.

13. The inflatable boat cover device of claim 10, wherein said plurality of fasteners are a second series of interior fasteners for securing said inflatable boat cover to the boat.

14. The inflatable boat cover device of claim 10, wherein said body component having a selectively openable door.

15. The inflatable boat cover device of claim 14, wherein said selectively openable door having a zipper for selectively fastening and unfastening said selectively openable door.

16. The inflatable boat cover device of claim 10, wherein said continuous air system is a portable air compressor.

17. A method of covering a boat with an inflatable boat cover, the method comprising the steps of: providing a body component, a continuous air system, a plurality of inflatable air tubes, a frame, and a plurality of fasteners, wherein each of said plurality of inflatable air tubes having a closeable threaded valve for inflation and deflation, further wherein said frame comprising said plurality of inflatable air tubes and said continuous air system for supporting said body component; securing

said inflatable boat cover to a boat with said plurality of fasteners including a first series of exterior fasteners; spanning each of said plurality of inflatable air tubes across an outer side of said body component from a first edge to an opposing second edge; and inflating said plurality of inflatable air tubes with said continuous air system.

18. The method of covering a boat with an inflatable boat cover of claim 17, wherein said body component having a material selected from the group consisting of a marine grade vinyl material, and further wherein said material of said body component is mildew resistant.

19. The method of covering a boat with an inflatable boat cover of claim 17, wherein said plurality of inflatable air tubes are interlocking, and further wherein each of said plurality of inflatable air tubes are selectively removable.

20. The method of covering a boat with an inflatable boat cover of claim 19, wherein said body component having a selectively openable door, and further wherein said selectively openable door having a zipper for selectively fastening and unfastening said selectively openable door.
